Technical University of Denmark

Time Series Analysis 02417

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Assignment 2

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1 Stability

We consider the AR(2) process

$$X_t + \phi_1 X_{t-1} + \phi_2 X_{t-2} = \varepsilon_t, \qquad \varepsilon_t \sim \mathrm{WN}(0,1), \tag{1}$$

with

$$\phi_1 = -0.7, \qquad \phi_2 = -0.2.$$

In the more familiar AR notation, this can be written as

$$X_t = 0.7X_{t-1} + 0.2X_{t-2} + \varepsilon_t.$$

1.1 Stationarity from the characteristic roots

For the process in (1), the characteristic polynomial is

$$\phi(z) = 1 + \phi_1 z + \phi_2 z^2.$$

Substituting the given values gives

$$1 - 0.7z - 0.2z^2 = 0,$$

or equivalently

$$0.2z^2 + 0.7z - 1 = 0.$$

Using the quadratic formula,

$$z = \frac{-0.7 \pm \sqrt{0.7^2 - 4(0.2)(-1)}}{2(0.2)} = \frac{-0.7 \pm \sqrt{1.29}}{0.4}.$$

Hence the two roots are approximately

$$z_1 \approx 1.09, \qquad z_2 \approx -4.59.$$

Both roots satisfy

$$|z_1| > 1, \qquad |z_2| > 1,$$

so both roots lie outside the unit circle. Therefore, the AR(2) process is **stationary**.

1.2 Invertibility

Invertibility is mainly a property of moving-average models. Since the present model is a pure AR(2) process, the relevant condition is stationarity rather than invertibility. Thus, the key conclusion here is that the process is stationary.

1.3 Autocorrelation function

For the AR(2) process, the autocorrelation function satisfies the Yule-Walker recursion

$$\rho(k) + \phi_1 \rho(k-1) + \phi_2 \rho(k-2) = 0, \qquad k \geq 2.$$

Also,

$$\rho(0) = 1.$$

At lag $k = 1$,

$$\rho(1) + \phi_1 \rho(0) + \phi_2 \rho(1) = 0,$$

which implies

$$\rho(1) = \frac{-\phi_1}{1+\phi_2}.$$

Therefore, for this AR(2) model,

$$\rho(0) = 1, \qquad \rho(1) = \frac{-\phi_1}{1+\phi_2}, \qquad \rho(k) = -\phi_1\rho(k-1) - \phi_2\rho(k-2), \quad k \geq 2.$$

With $\phi_1 = -0.7$ and $\phi_2 = -0.2$,

$$\rho(1) = \frac{0.7}{0.8} = 0.875,$$

and then

$$\rho(2) = 0.8125, \qquad \rho(3) = 0.74375, \qquad \rho(4) = 0.683125.$$

Thus the autocorrelations are positive and decay gradually toward zero.

1.4 Plot of the autocorrelation function

Figure 1 shows the theoretical ACF up to lag 30. The plot confirms the expected AR behavior: the ACF is positive at short lags and decreases gradually rather than cutting off abruptly.

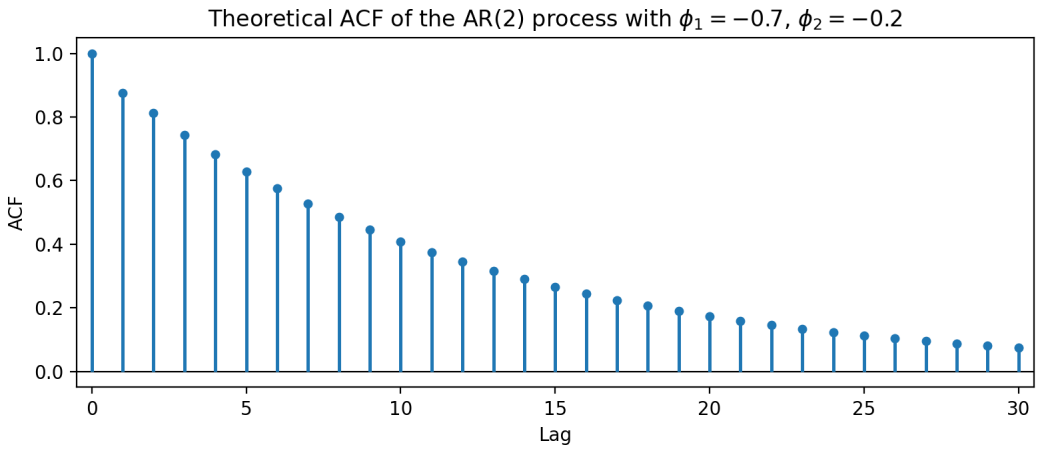


Figure 1: Theoretical ACF up to lag 30 for the AR(2) process with $\phi_1 = -0.7$ and $\phi_2 = -0.2$.

2 Simulating seasonal processes

In this section, the seasonal period is $s = 12$, and all processes are simulated in Python with a fixed seed for reproducibility. For each model, a time-series plot together with its ACF and PACF is shown. The key idea is to distinguish between:

- **non-seasonal dependence**, which acts mainly at lags $1, 2, 3, \ldots$,
- **seasonal dependence**, which acts mainly at lags $12, 24, 36, \ldots$.

2.1 $(1,0,0) \times (0,0,0)_{12}$ with $\phi_1 = 0.6$

This is a simple non-seasonal AR(1) model:

$$Y_t = 0.6Y_{t-1} + \varepsilon_t.$$

The process shows short-run persistence but no seasonal structure. In the ACF, the correlations decay gradually from lag 1 onward, which is the standard AR(1) pattern. The PACF shows one dominant spike at lag 1 and is close to zero afterwards.

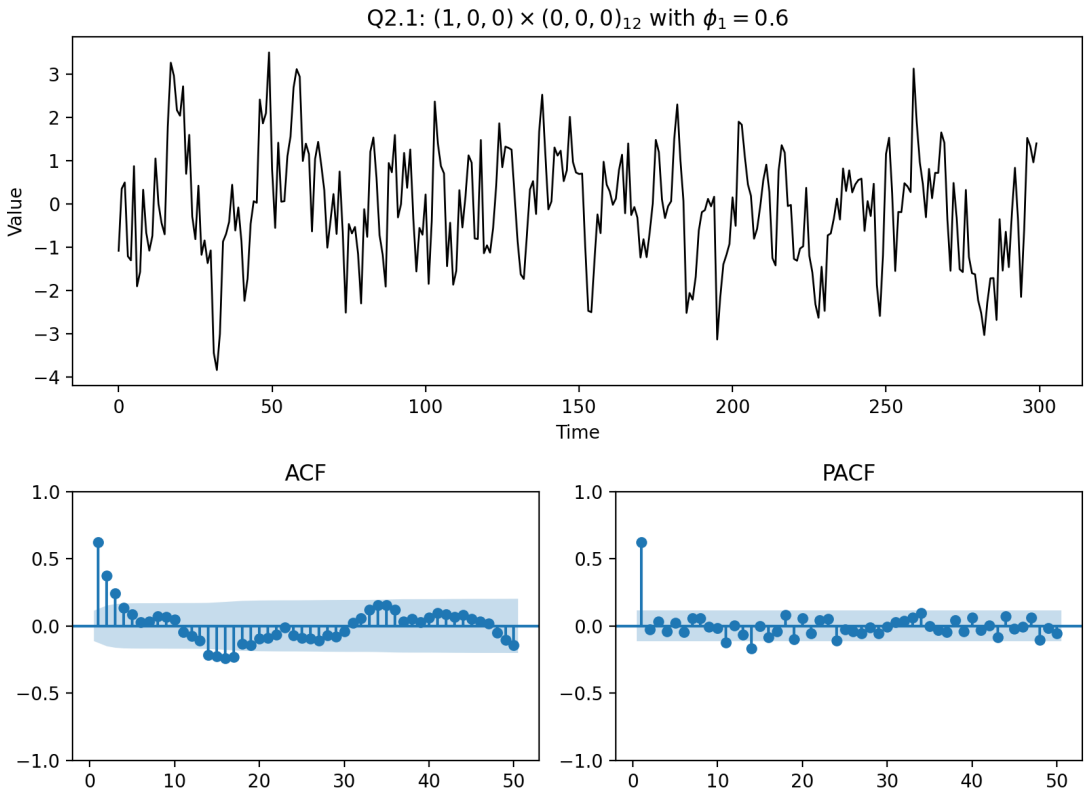


Figure 2: Simulation, ACF and PACF for $(1,0,0) \times (0,0,0)_{12}$ with $\phi_1 = 0.6$.

2.2 $(0,0,0) \times (1,0,0)_{12}$ with $\Phi_1 = -0.9$

This is a seasonal AR(1) model:

$$Y_t = -0.9Y_{t-12} + \varepsilon_t.$$

The dependence is therefore seasonal rather than short-run. The ACF shows pronounced seasonal spikes at lags $12, 24, 36, \ldots$. Because the seasonal coefficient is negative, the signs alternate across seasonal lags. The PACF shows a dominant spike at lag 12, which is the characteristic footprint of a seasonal AR term.

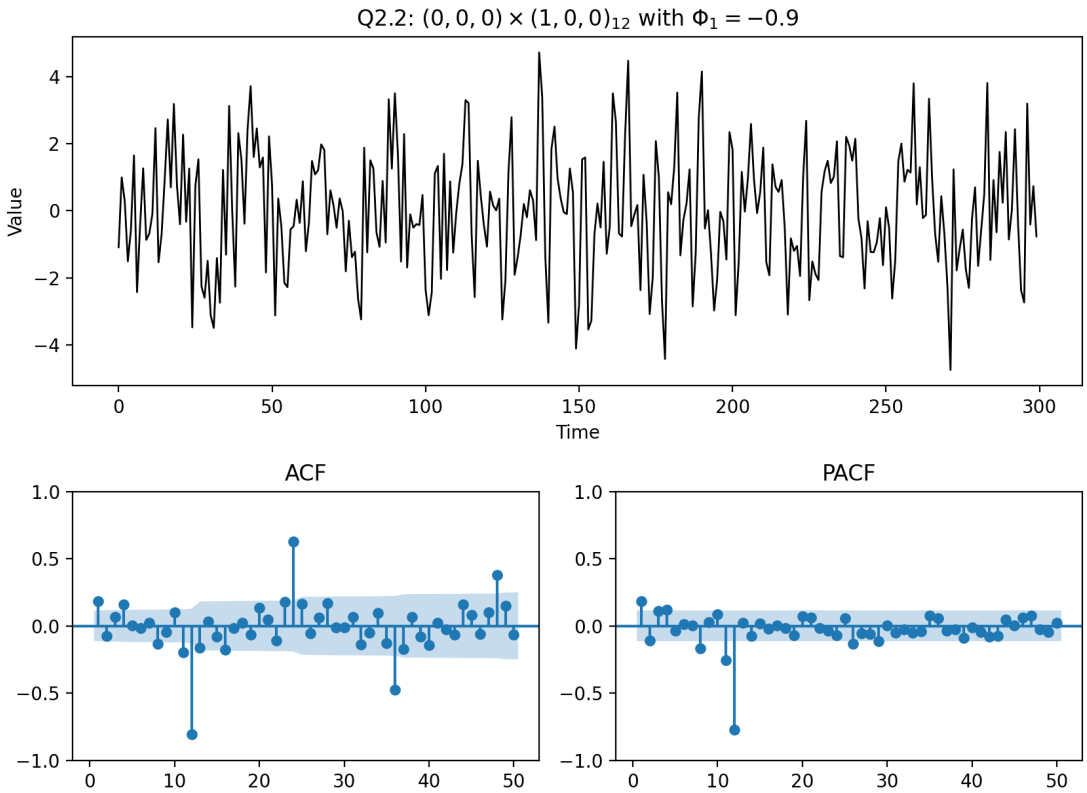


Figure 3: Simulation, ACF and PACF for $(0,0,0) \times (1,0,0)_{12}$ with $\Phi_1 = -0.9$.

2.3 $(1,0,0) \times (0,0,1)_{12}$ with $\phi_1 = 0.9$ and $\Theta_1 = -0.7$

This model combines a strong non-seasonal AR(1) component with a seasonal MA(1) component:

$$(1 - 0.9B)Y_t = (1 - 0.7B^{12})\varepsilon_t.$$

Equivalently,

$$Y_t = 0.9Y_{t-1} + \varepsilon_t - 0.7\varepsilon_{t-12}.$$

The AR coefficient is large, so the series is highly persistent and the ACF decays slowly. The seasonal MA term introduces a visible seasonal feature around lag 12. In the PACF, the strongest spike is at lag 1, while some additional seasonal structure may appear near lag 12.

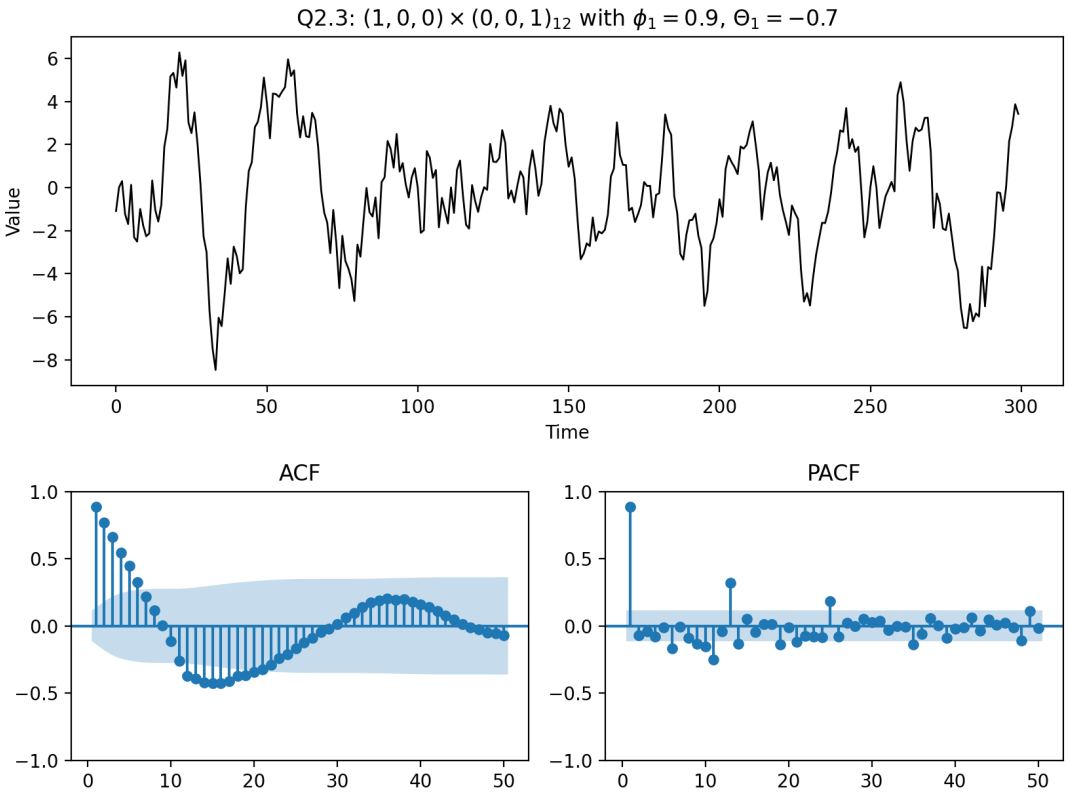


Figure 4: Simulation, ACF and PACF for $(1,0,0)\times(0,0,1)_{12}$ with $\phi_1 = 0.9$ and $\Theta_1 = -0.7$.

2.4 $(1,0,0)\times(1,0,0)_{12}$ with $\phi_1 = -0.6$ and $\Phi_1 = -0.8$

This model contains both a non-seasonal AR(1) term and a seasonal AR(1) term:

$$(1+0.6B)(1+0.8B^{12})Y_t = \varepsilon_t.$$

Hence the process has both short-run and seasonal autoregressive dependence. Because both coefficients are negative, the series tends to exhibit alternating or oscillatory behavior. The ACF tails off rather than cutting off, but it also shows clear seasonal structure at the multiples of 12. The PACF is expected to show important spikes at lag 1 and lag 12, and the multiplicative interaction may also create a visible effect around lag 13.

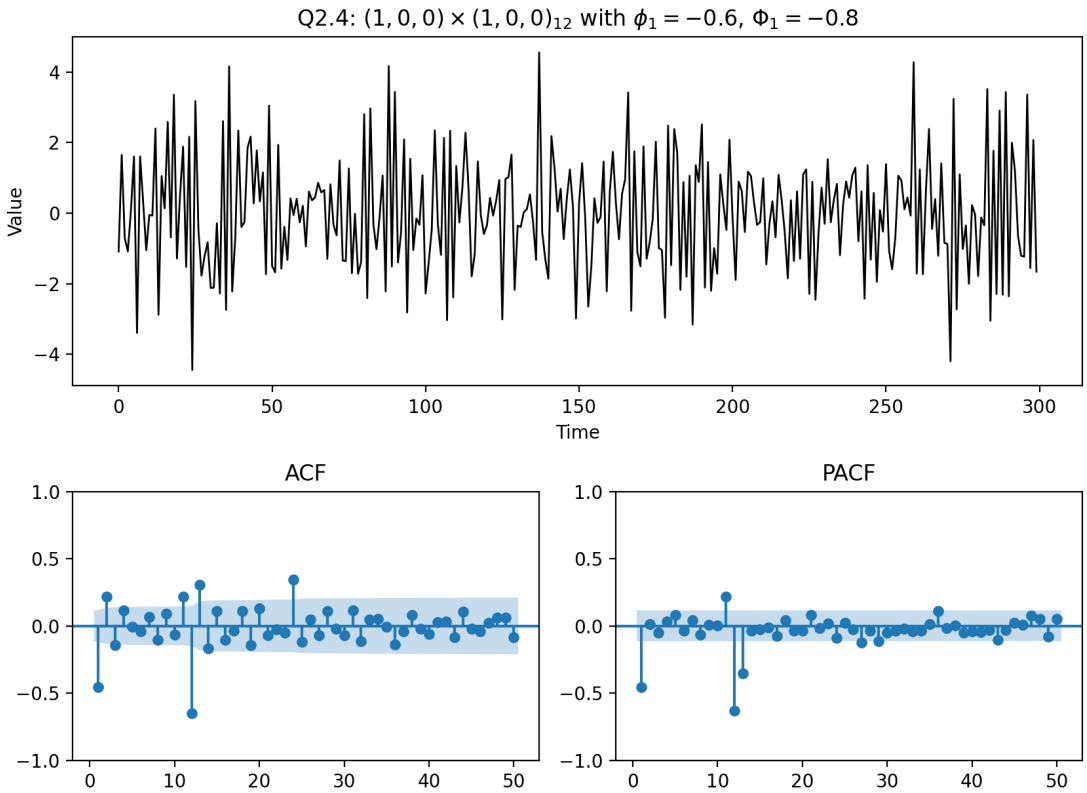


Figure 5: Simulation, ACF and PACF for $(1,0,0)\times(1,0,0)_{12}$ with $\phi_1=-0.6$ and $\Phi_1=-0.8$.

2.5 $(0,0,1)\times(0,0,1)_{12}$ with $\theta_1=0.4$ and $\Theta_1=-0.8$

This model contains both a non-seasonal MA(1) term and a seasonal MA(1) term:

$$Y_t=(1+0.4B)(1-0.8B^{12})\varepsilon_t.$$

Expanding the product gives dependence on ε_t, ε_{t-1}, ε_{t-12}, and ε_{t-13}. Therefore the ACF is expected to show a small number of important spikes, especially around lags 1, 12 and 13, after which it should quickly become negligible. The PACF is not expected to cut off sharply and instead decays more gradually.

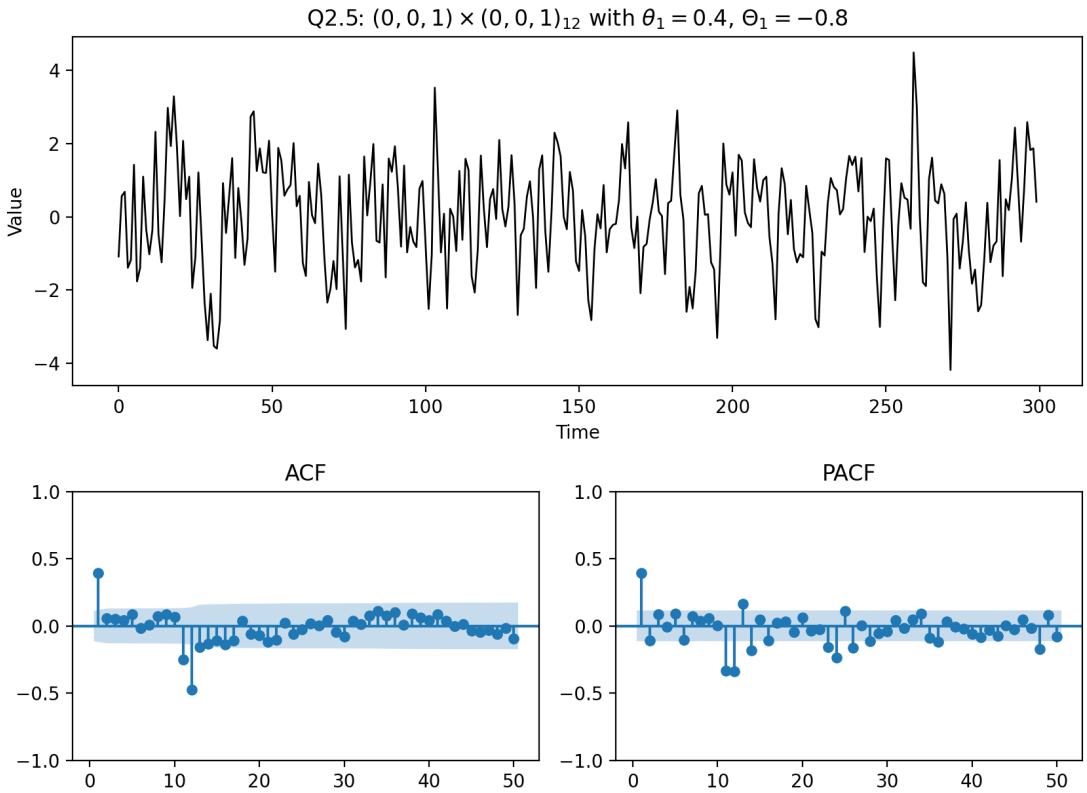


Figure 6: Simulation, ACF and PACF for $(0,0,1)\times(0,0,1)_{12}$ with $\theta_1 = 0.4$ and $\Theta_1 = -0.8$.

2.6 $(0,0,1)\times(1,0,0)_{12}$ with $\theta_1 = -0.4$ and $\Phi_1 = 0.7$

This model combines a non-seasonal MA(1) effect with a seasonal AR(1) effect:

$$(1 - 0.7B^{12})Y_t = (1 - 0.4B)\varepsilon_t.$$

Equivalently,

$$Y_t = 0.7Y_{t-12} + \varepsilon_t - 0.4\varepsilon_{t-1}.$$

The non-seasonal MA part gives a short-run effect, mostly visible at lag 1. The seasonal AR part generates persistence at seasonal lags $12, 24, 36, \ldots$. Consequently, the ACF shows a mixed pattern: a short-lag MA feature together with gradually decaying seasonal spikes. The PACF again highlights the seasonal structure, especially around lag 12.

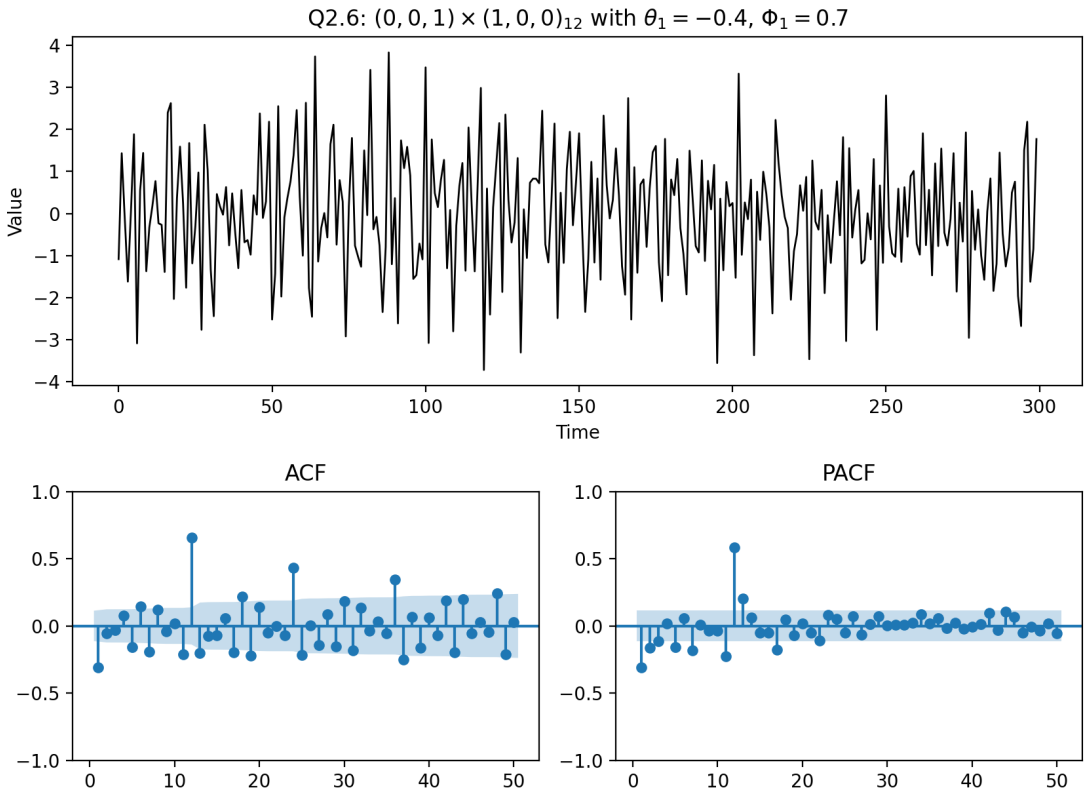


Figure 7: Simulation, ACF and PACF for $(0,0,1)\times(1,0,0)_{12}$ with $\theta_1=-0.4$ and $\Phi_1=0.7$.

2.7 Summary of observations

The main conclusions from the simulated seasonal models are as follows:

- Non-seasonal AR terms create gradual decay in the ACF at short lags.
- Non-seasonal MA terms create cutoff-type behavior in the ACF at short lags.
- Seasonal AR terms create recurring ACF spikes at the seasonal lags $12, 24, 36, \ldots$, typically with gradual decay across those seasonal multiples.
- Seasonal MA terms create more limited seasonal effects, often concentrated around a small number of seasonal lags.
- In mixed seasonal models, the ACF and PACF combine short-run and seasonal information, so neither function is always simple to interpret in isolation.

Table 1 summarizes the qualitative behavior.

Table 1: Qualitative ACF/PACF behavior for the simulated seasonal models.

Model	ACF behavior	PACF behavior
$(1,0,0) \times (0,0,0)_{12}$	Gradual decay at short lags	Dominant lag-1 spike
$(0,0,0) \times (1,0,0)_{12}$	Seasonal spikes at $12, 24, 36, \ldots$ with alternating sign	Dominant lag-12 spike
$(1,0,0) \times (0,0,1)_{12}$	Slow decay plus seasonal feature near lag 12	Strong lag-1 spike, possible seasonal structure
$(1,0,0) \times (1,0,0)_{12}$	Decaying short-run and seasonal oscillation	Spikes near lags 1 and 12
$(0,0,1) \times (0,0,1)_{12}$	Important spikes near lags 1, 12 and 13, then cutoff	Gradual decay
$(0,0,1) \times (1,0,0)_{12}$	Short-lag MA feature plus seasonal AR decay	Seasonal structure near lag 12

In general, the most important lesson is that seasonal dependence should be looked for at the seasonal lags rather than only at lags $1, 2, 3, \ldots$. For monthly data with period 12, the lags $12, 24, 36, \ldots$ are often the most informative for identifying seasonal structure.

3 Identifying ARMA models

In this section, three simulated processes are identified from their time-series plots, ACFs and PACFs. Since the problem explicitly asks for a guess, the emphasis is on matching the plots to the standard qualitative signatures of AR, MA and ARMA models.

3.1 Process 1

Figure 8 shows a series with no clear serial persistence. Both the ACF and PACF stay close to zero for almost all lags, with only random fluctuations inside or near the confidence bounds. The most plausible identification is therefore

$$\boxed{\text{ARMA}(0,0)},$$

that is, white noise.

3.1. Process 1:

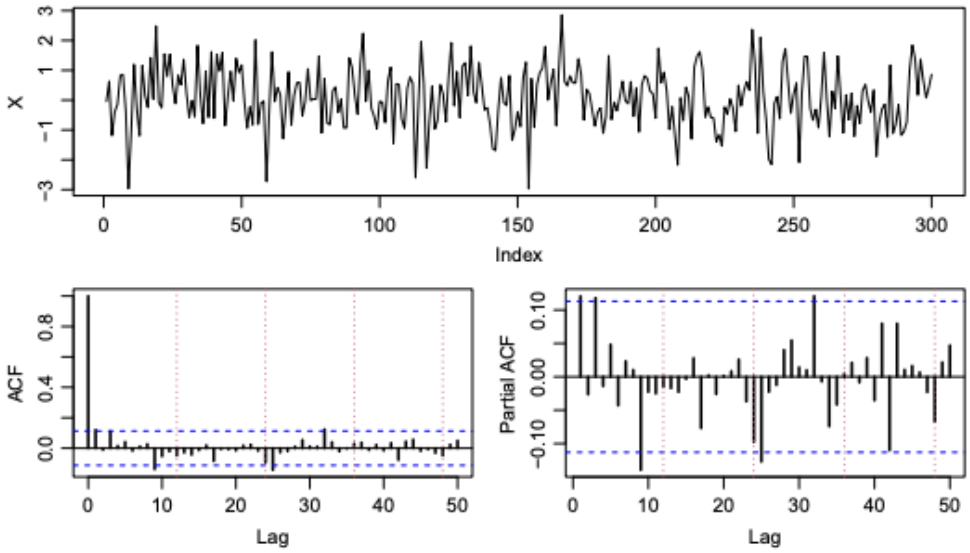


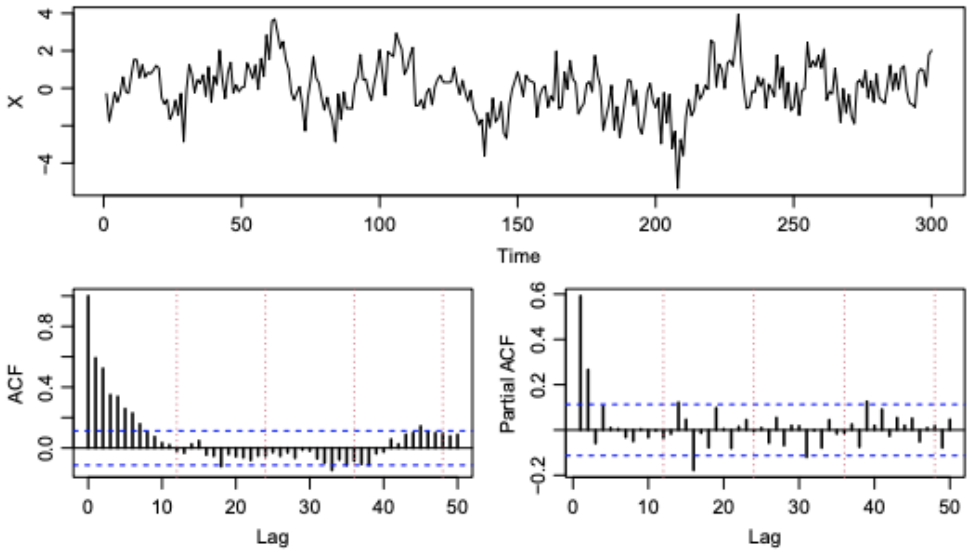
Figure 8: Process 1: time-series plot, ACF and PACF.

3.2 Process 2

Figure 9 shows an ACF that decays gradually rather than cutting off, while the PACF has two pronounced spikes at the first two lags and then becomes small. This is the classic signature of an AR(2) model, so the most plausible identification is

$$\boxed{\text{ARMA}(2,0)}.$$

3.2. Process 2:



3.3. Process 3:

Figure 9: Process 2: time-series plot, ACF and PACF.

3.3 Process 3

Figure 10 shows that neither the ACF nor the PACF cuts off cleanly. Instead, both functions decay gradually. This suggests a mixed ARMA structure rather than a pure AR or pure MA model. A reasonable low-order guess is

$$\boxed{\text{ARMA}(1,1)}.$$

Since identification from plots alone is not exact, this should be interpreted as the most plausible simple model rather than a unique answer.

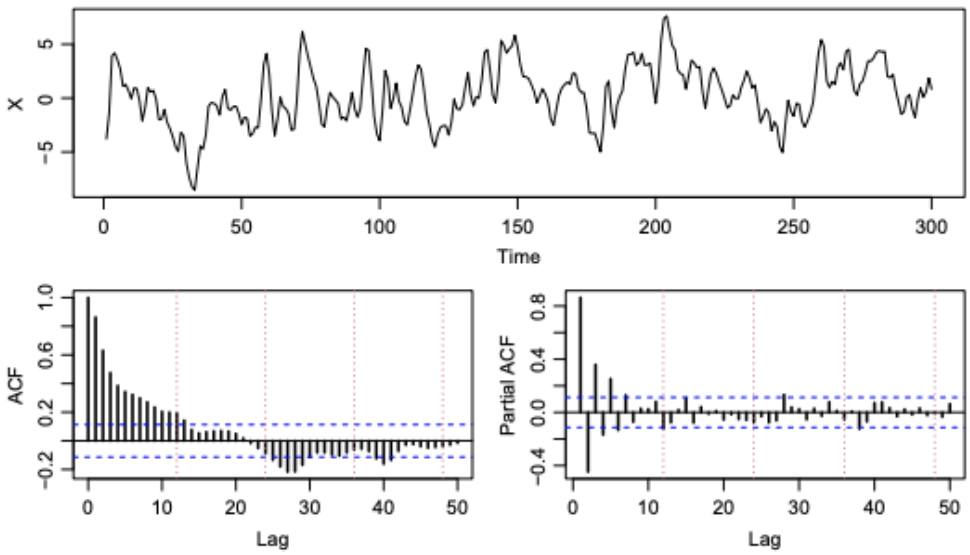


Figure 10: Process 3: time-series plot, ACF and PACF.

3.4 Summary

The guessed model structures are collected in Table 2. The main identification rule used throughout is:

- **AR model:** ACF tails off, PACF cuts off.
- **MA model:** ACF cuts off, PACF tails off.
- **ARMA model:** both ACF and PACF tail off.

Table 2: Guessed ARMA structures from the diagnostic plots.

Process	Guessed model	Reasoning
1	ARMA(0,0)	Both ACF and PACF are close to zero
2	ARMA(2,0)	ACF tails off, PACF has two main spikes
3	ARMA(1,1)	Both ACF and PACF tail off

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